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Clinical review of robotic distal pancreatectomy

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≈1 day shorter hospital stay than laparoscopic

0% positive margins (laparoscopic 2%)

≤ LDP overall cost; well below open surgery

+30 min operating time vs laparoscopic

01 BACKGROUND & AIM

- Open (ODP) → laparoscopic (LDP) → robotic (RDP)
- LDP is limited by dexterity and visualisation
- Whether RDP improves outcomes is debated

AIM Compare RDP with ODP and LDP: technique, outcomes, cost, challenges

02 WHY ROBOTIC?

3D magnified vision

Tremor filtration

Motion scaling

Wristed instruments

Platforms: evidence is mostly da Vinci; Versius and hinotori™ have only case reports

03 OPERATIVE TECHNIQUE

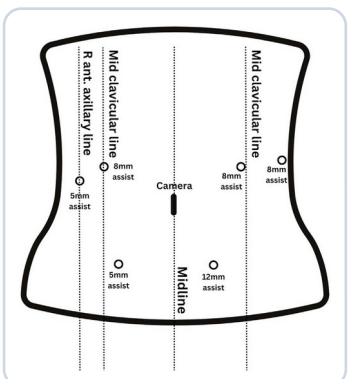


Fig. 1 Port placement

1 **Position** Supine; reverse Trendelenburg; dock

2 **Expose** Enter the lesser sac

3 **Control** Splenic artery and vein

4 **Divide** Pancreas with clear margin

5 **Resect** En bloc ± spleen

Spleen preserving: vessel preservation or Warshaw

04 CLINICAL OUTCOMES OF RDP

Outcome	RDP vs open	RDP vs laparoscopic
Blood loss	▼ Lower	▼ Lower (planned SP)
Complications	▼ Fewer (fistula, wound)	= Similar overall and fistula
Hospital stay	▼ Shorter	▼ ≈1 day shorter
Conversion to open	Not applicable	▼ Fewer
Spleen preservation	▲ Higher	▲ Higher; fewer splenectomies
Oncological	= Non-inferior R0 and nodes	▲ 0% vs 2% R1; more lymph nodes
Operative time	= Conflicting	▲ ≈30 min longer (docking)

■ favours RDP ■ no clear difference ■ favours comparator SP, spleen preservation

05 COST-EFFECTIVENESS

Operative cost **RDP higher**

Overall cost **RDP ≤ LDP < ODP**

Quality of life **RDP > LDP**

- Theatre cost offset by shorter stay
- More QALYs than LDP at similar cost

06 CHALLENGES

Operative time Docking adds time

Team & training Few training centres

Access & cost High upfront cost; unequal access

CONCLUSION

RDP offers greater precision, better outcomes and more spleen preservation than open and laparoscopic surgery, and is cost-effective. Wider adoption needs training and equitable access.